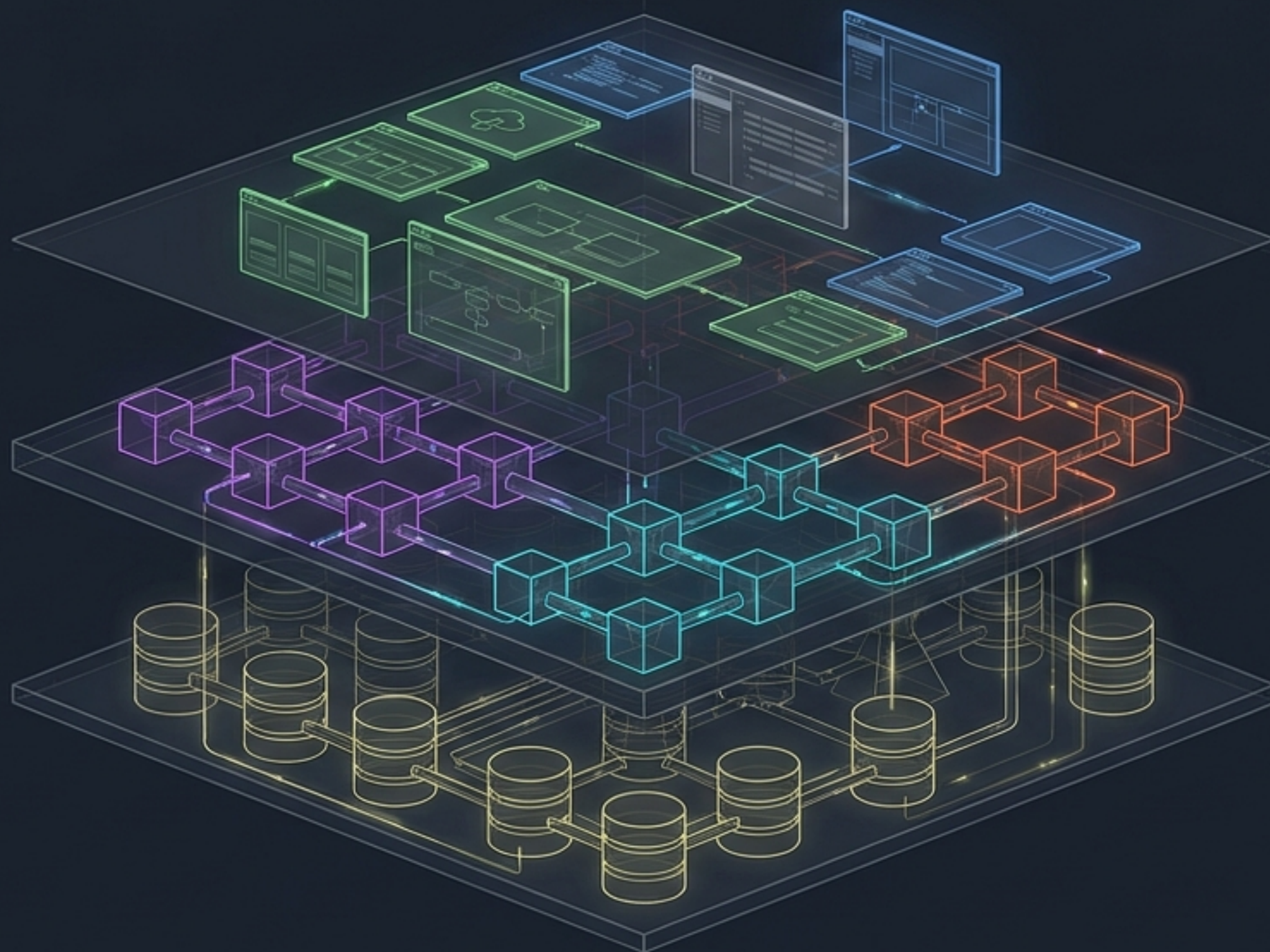


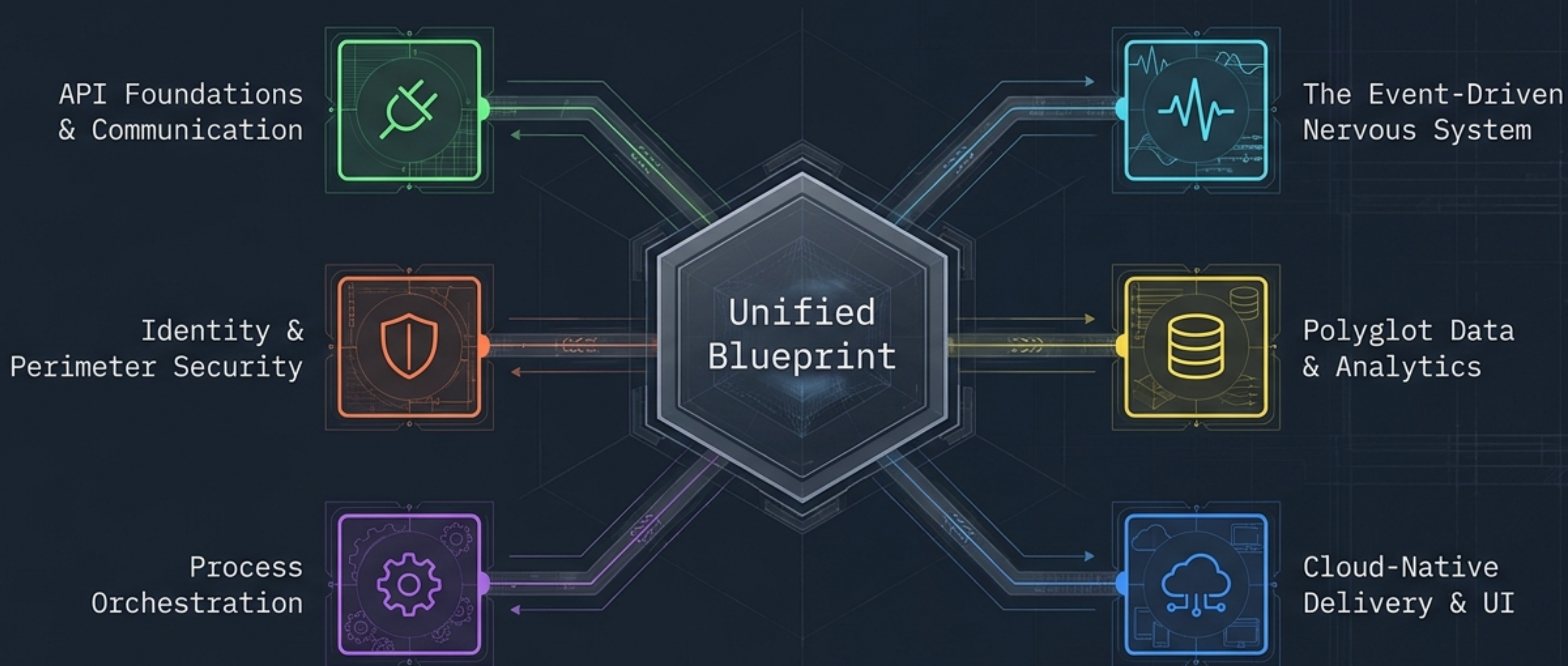
The Architecture Gallery

Constructing the Unified Enterprise Blueprint
A Synthesis of the k1729p Portfolio (Studies 01-28)



The Enterprise Anatomy

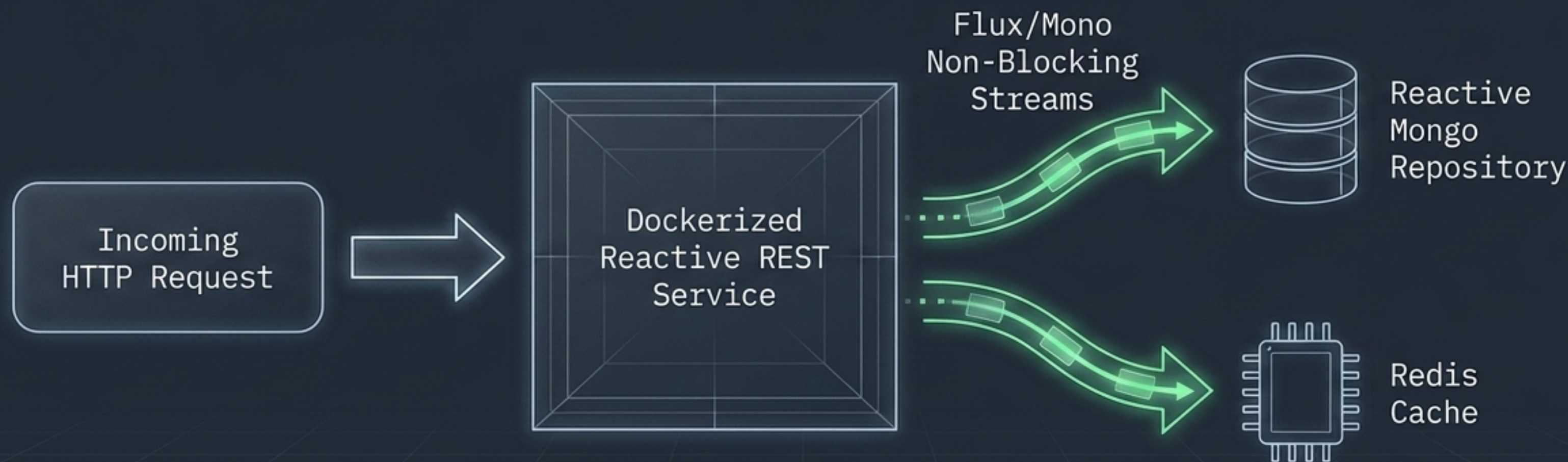
Mapping 28 independent technical studies into a single, cohesive, enterprise-grade architecture.



API Foundations & Reactive Systems



Implementing Spring WebFlux and OpenAPI. Bridging Functional Endpoints with Annotated Controllers to achieve non-blocking data flows.





Protocol Evolution: GraphQL vs. gRPC

Tailoring the communication protocol to the payload requirements.

Criteria	Study 25: GraphQL	Study 26: gRPC
Schema Enforcement	Entity graphs & @auth directives	HTTP/2 Protocol Buffers (.proto)
Payload Efficiency	Exact data fetching (No over-fetching)	Highly compressed binary data
Streaming Capability	Subscriptions	Bidirectional Streaming RPC
Architectural Use Case	Complex data dashboards	High-velocity Chat Services

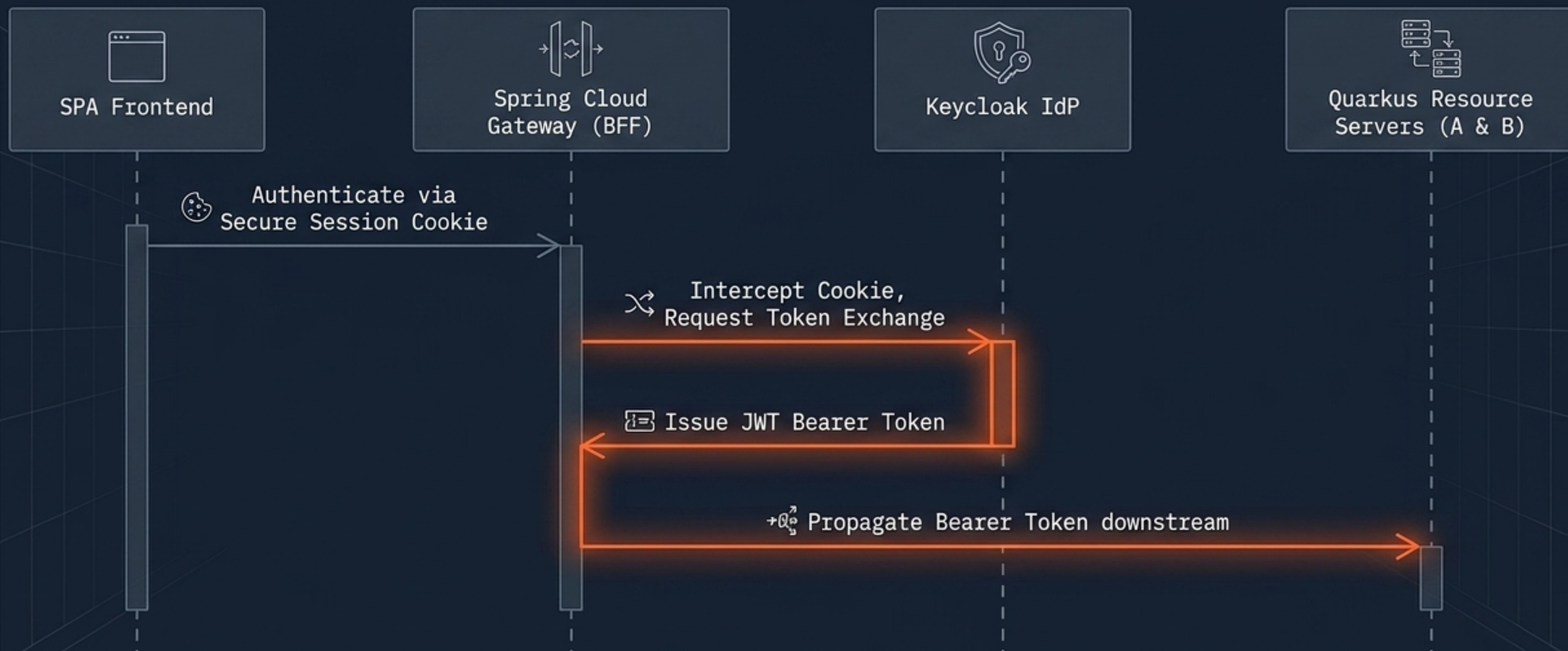
Client

Read-Write Stream

Server

The BFF Security Perimeter

Securing the blueprint with Keycloak, Spring Cloud Gateway, and Quarkus using Role-Based Access Control (RBAC).



Executable Business Orchestration

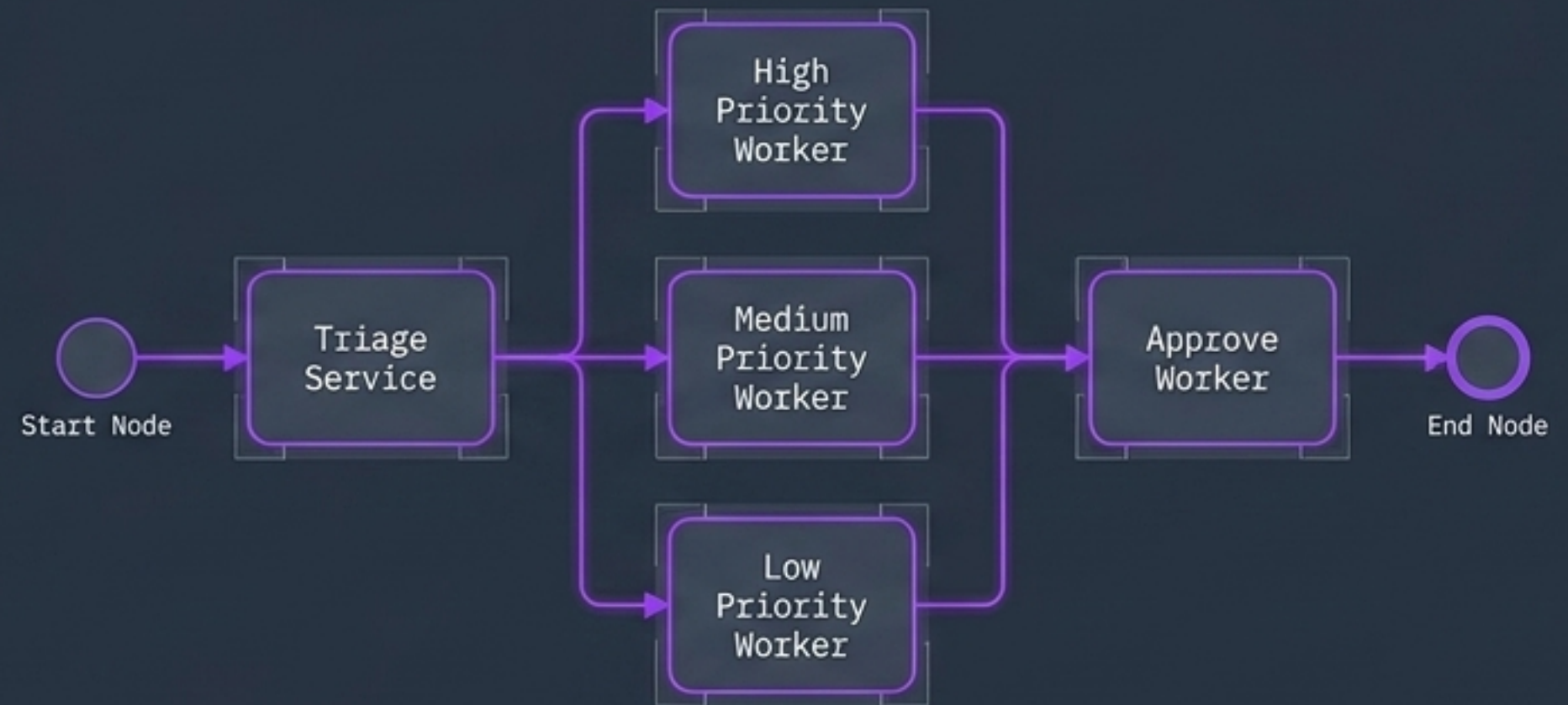
Translating Decision Requirements Diagrams (DRD) into Zeebe worker execution via Camunda Platform.



The Logic (DMN)

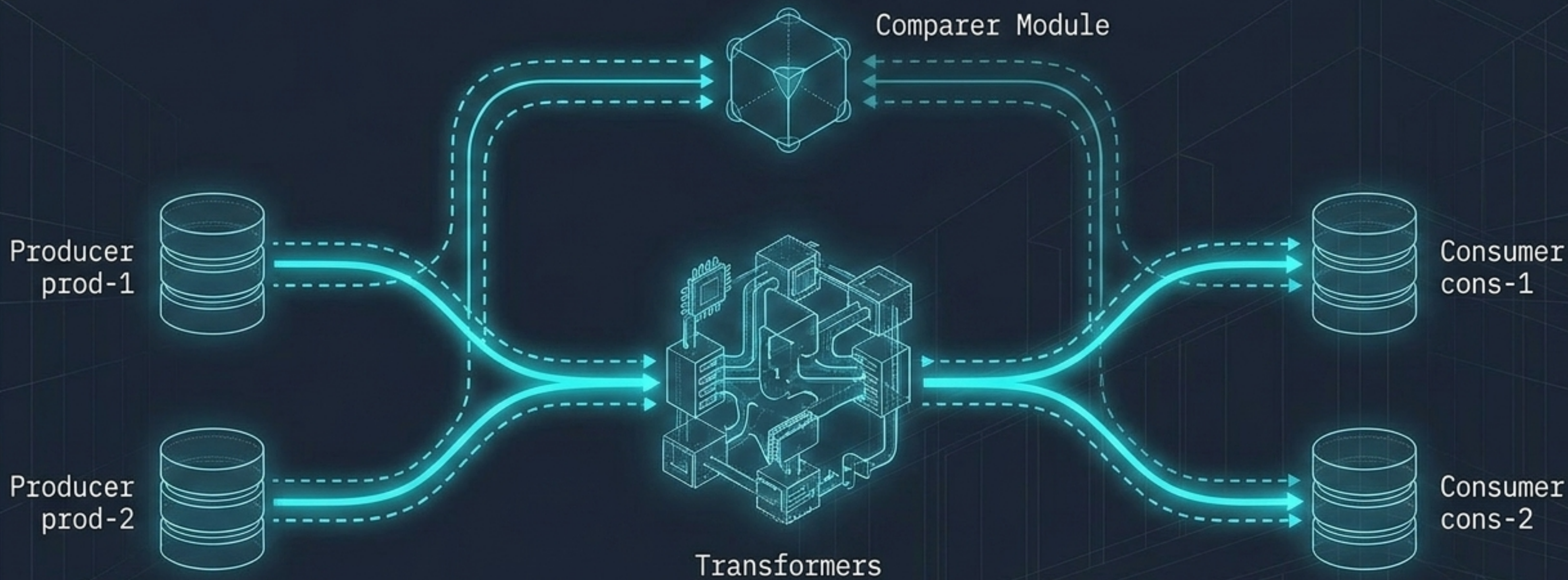
	Priority Score	Impact (High, Med, Low)		
		High	Med	Low
Urgency (High, Med, lw)	High	100	70	30
	Med	70	50	20
	Low	30	20	10

The Execution (BPMN)



The Event-Driven Nervous System

Decoupling microservices and processing real-time topologies via Kafka Brokers and Spring Cloud Stream Binders.

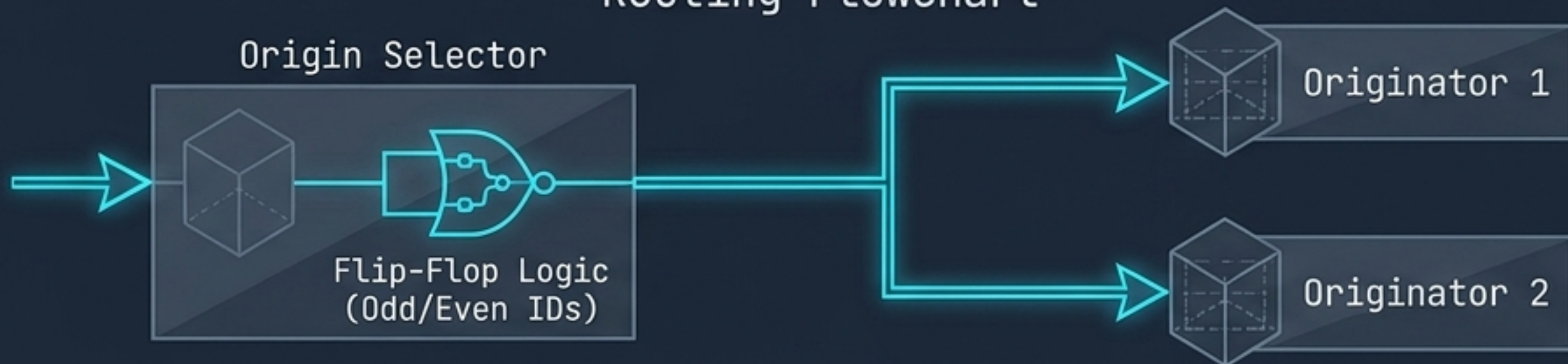


Advanced Failover & Message Routing

Ensuring high availability using Apache Pulsar's active/standby consumer subscriptions and flip-flop routing logic.



Routing Flowchart

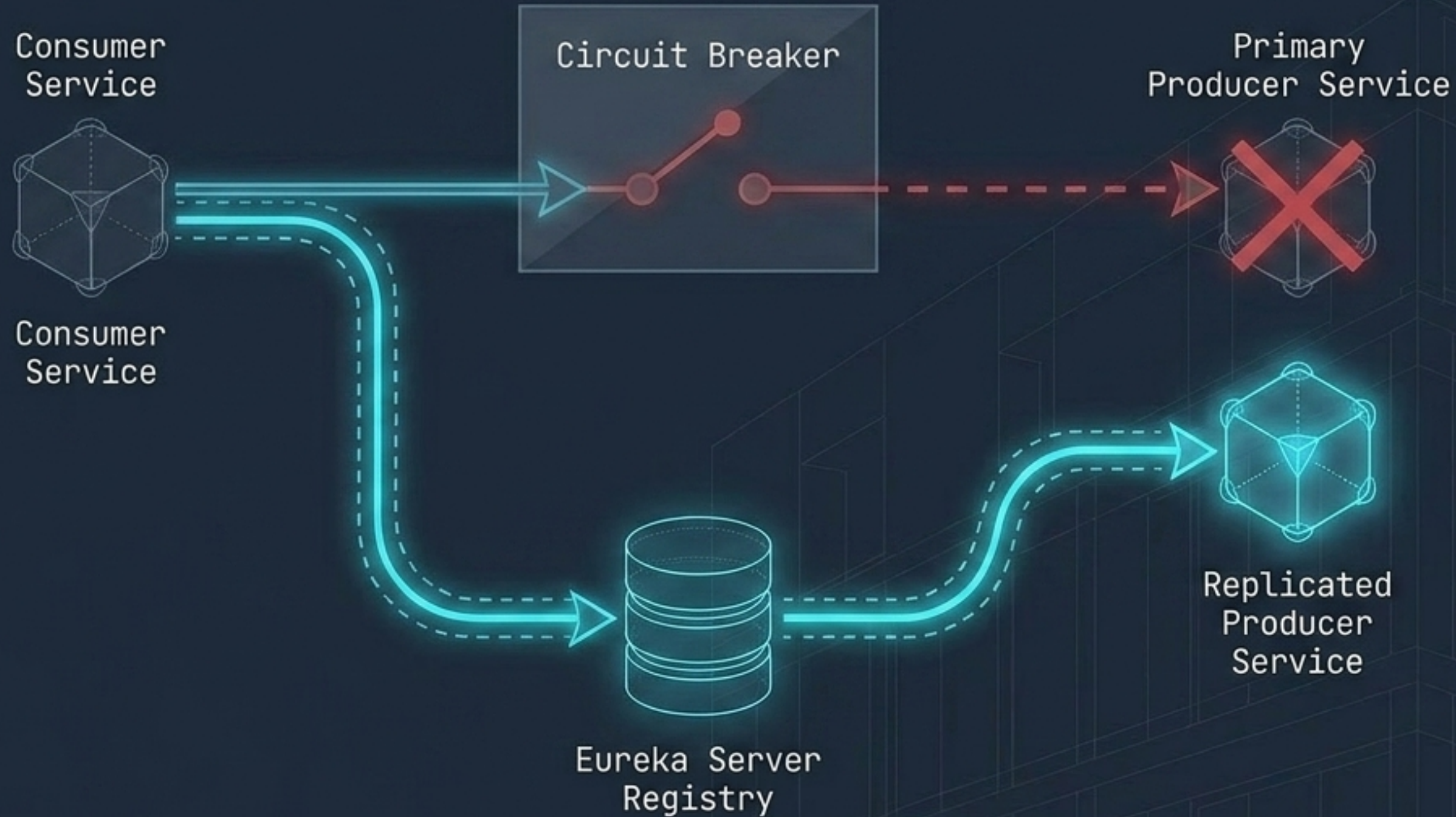


Pulsar Fail-over Subscription



Microservice Discovery & Resilience

Preventing cascading failures across the enterprise using Spring Cloud Netflix Eureka and Resilience4j.

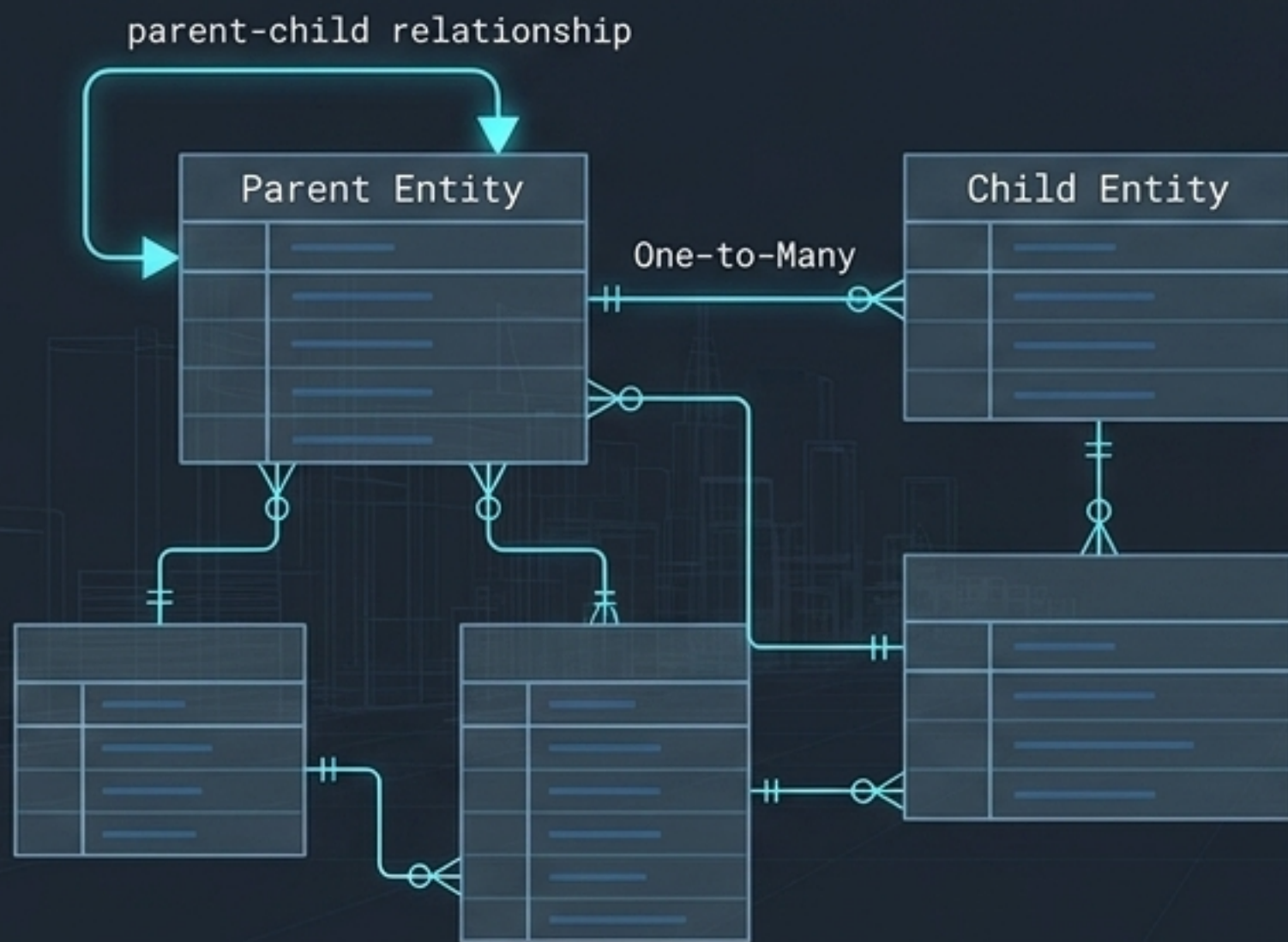


Legacy Integration & Enterprise Java

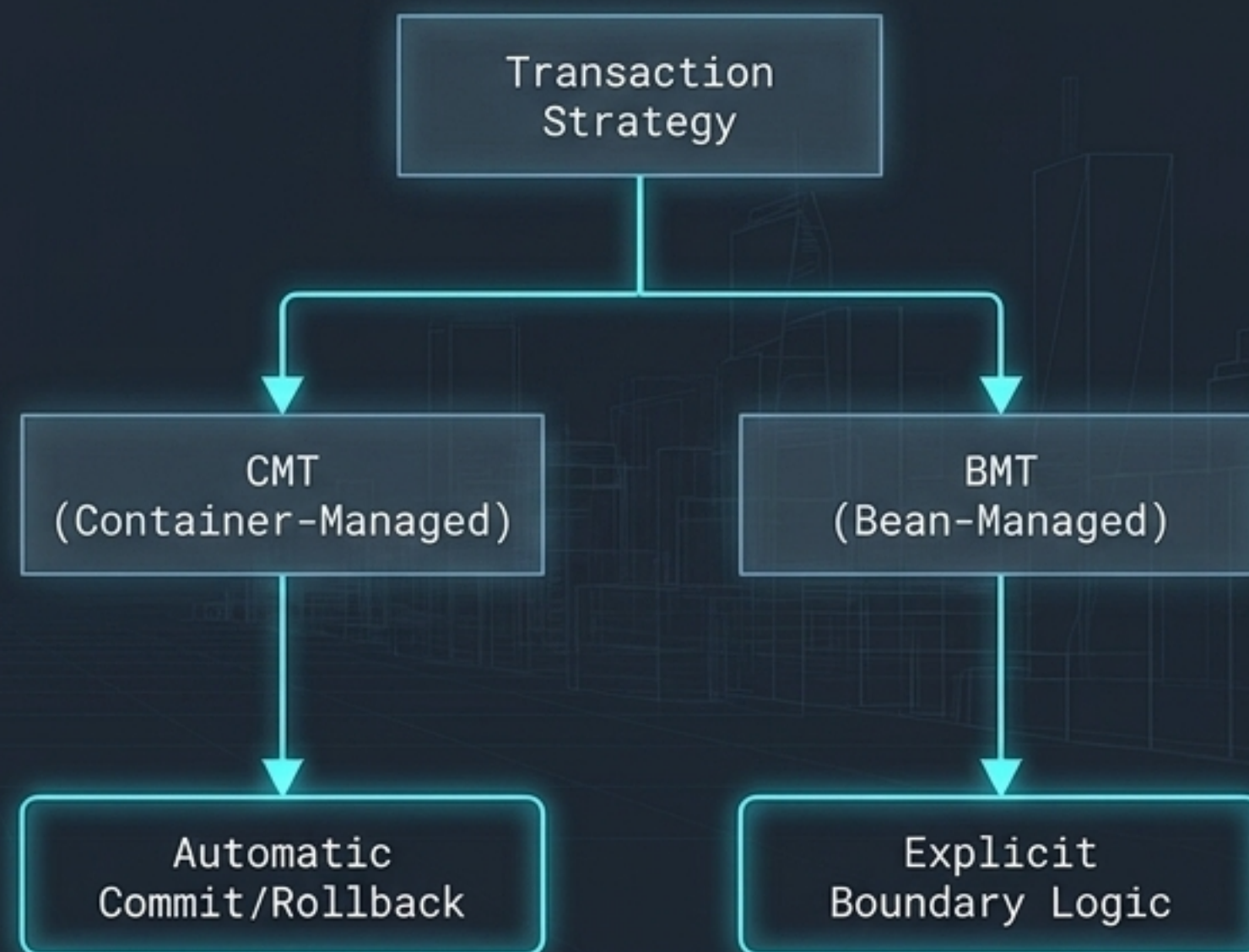
Bridging deep, complex legacy systems with modern infrastructure using JBoss WildFly, EJB, JPA, and SOAP.



JPA Entity Relationships



Transaction Management



Polyglot Persistence & Advanced Search

Ingesting, storing, and analyzing flat files into high-performance inverted-index clusters and multi-DB architectures.

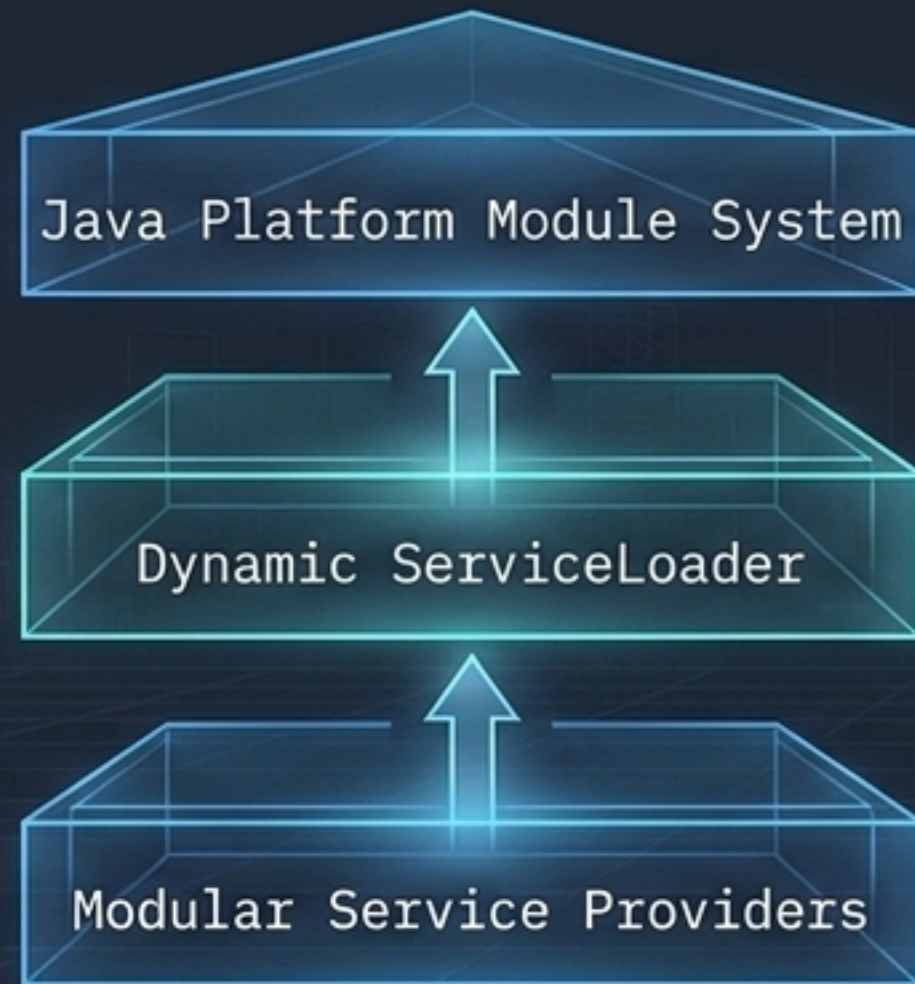


Core Performance & Algorithm Mastery

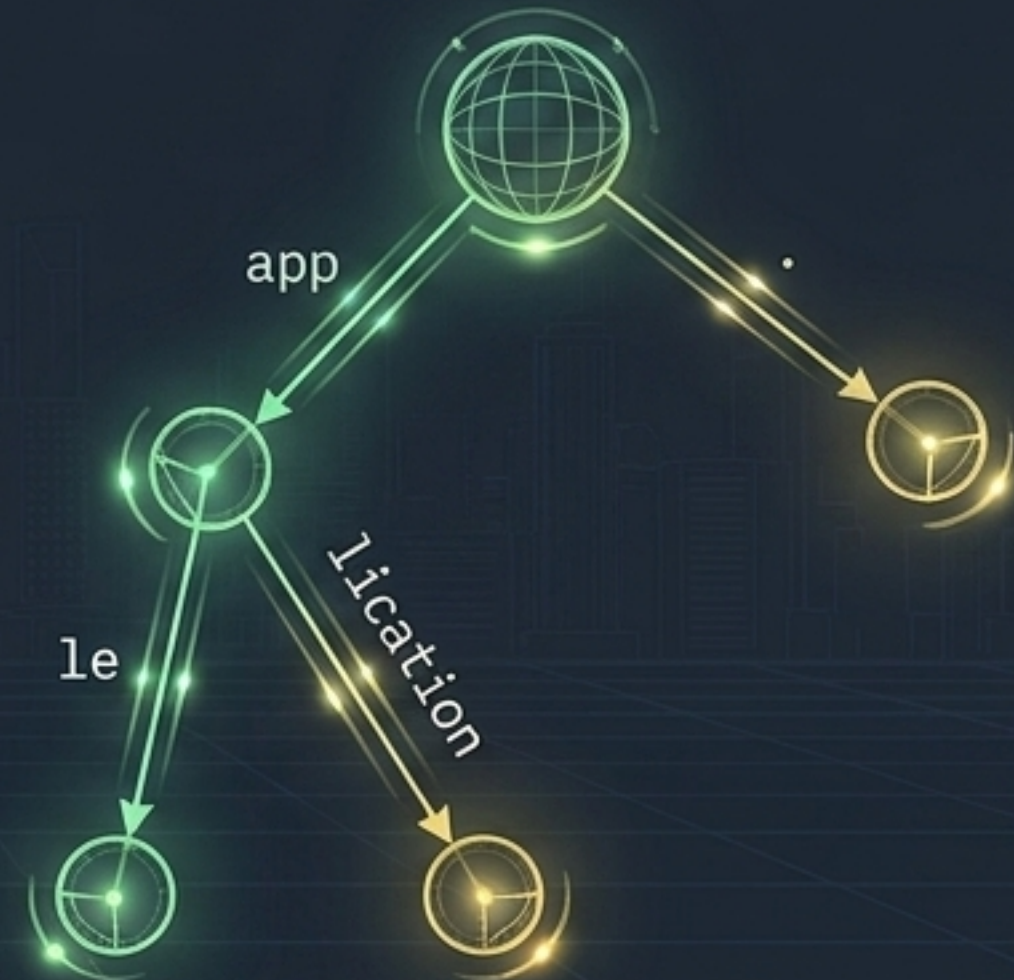
The bedrock of the blueprint: optimized Java code, Concurrent Trees, and Elliptic-Curve Cryptography.



Java Platform Module System

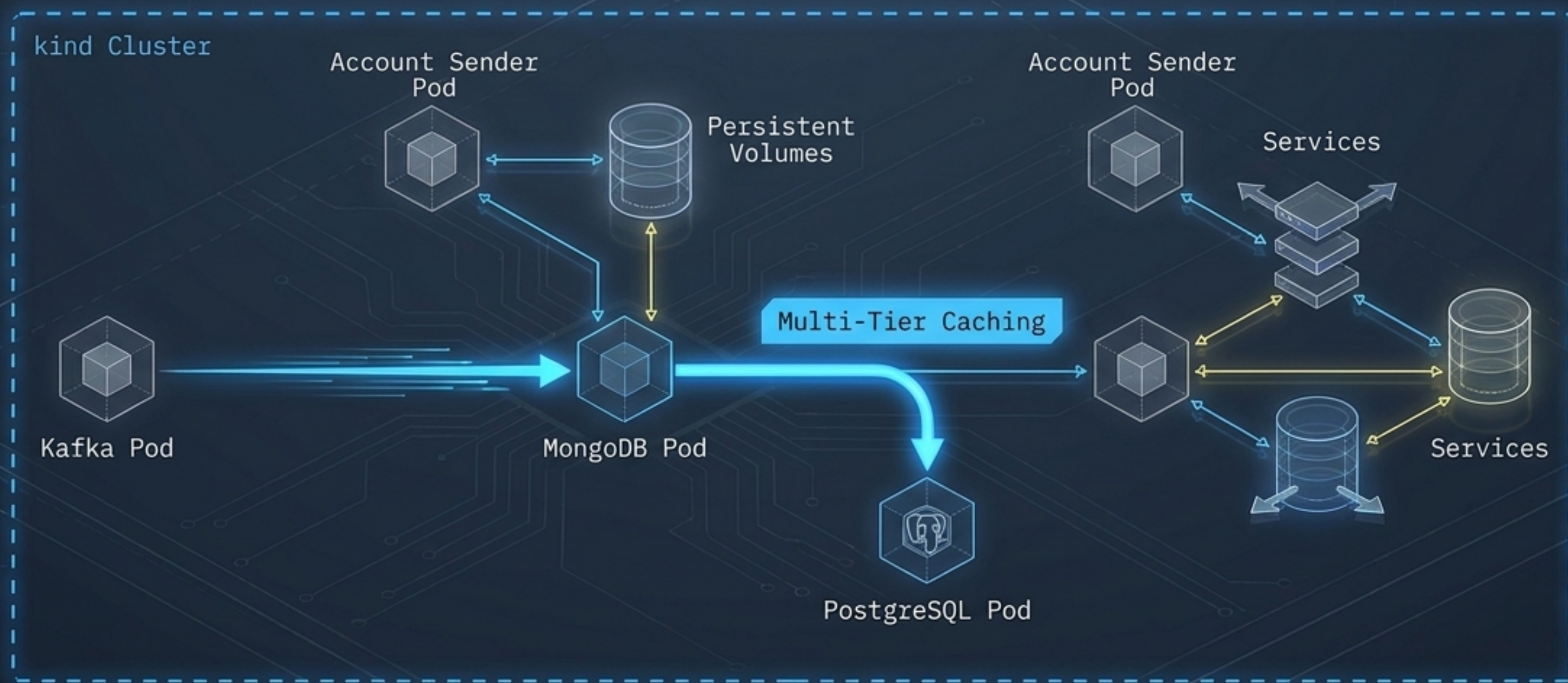


Patricia Trie Data Structure



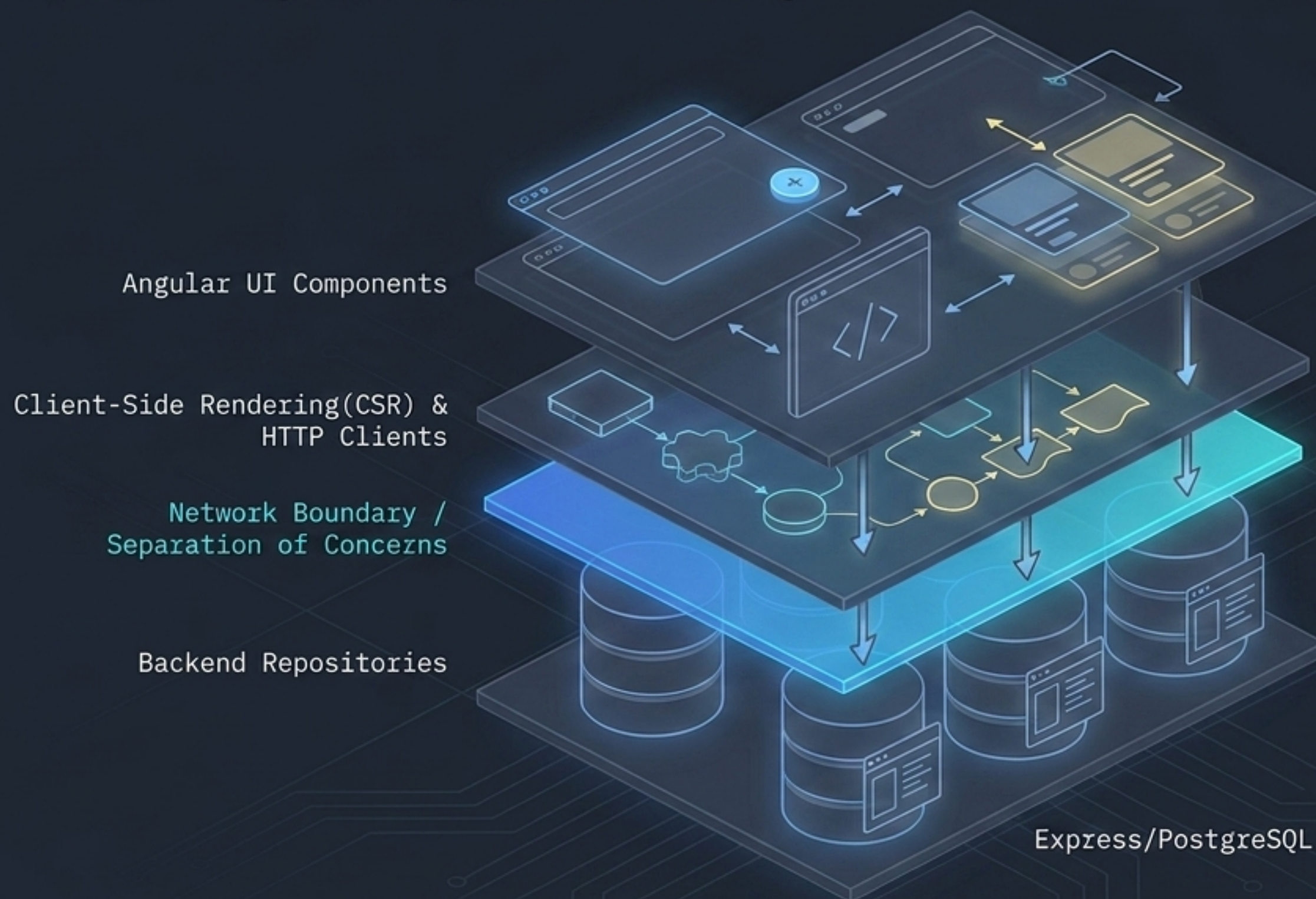
Cloud-Native Kubernetes Deployment

Packaging the enterprise. Deploying native Quarkus executables via Helm onto a kind cluster with a multi-tiered data caching strategy.



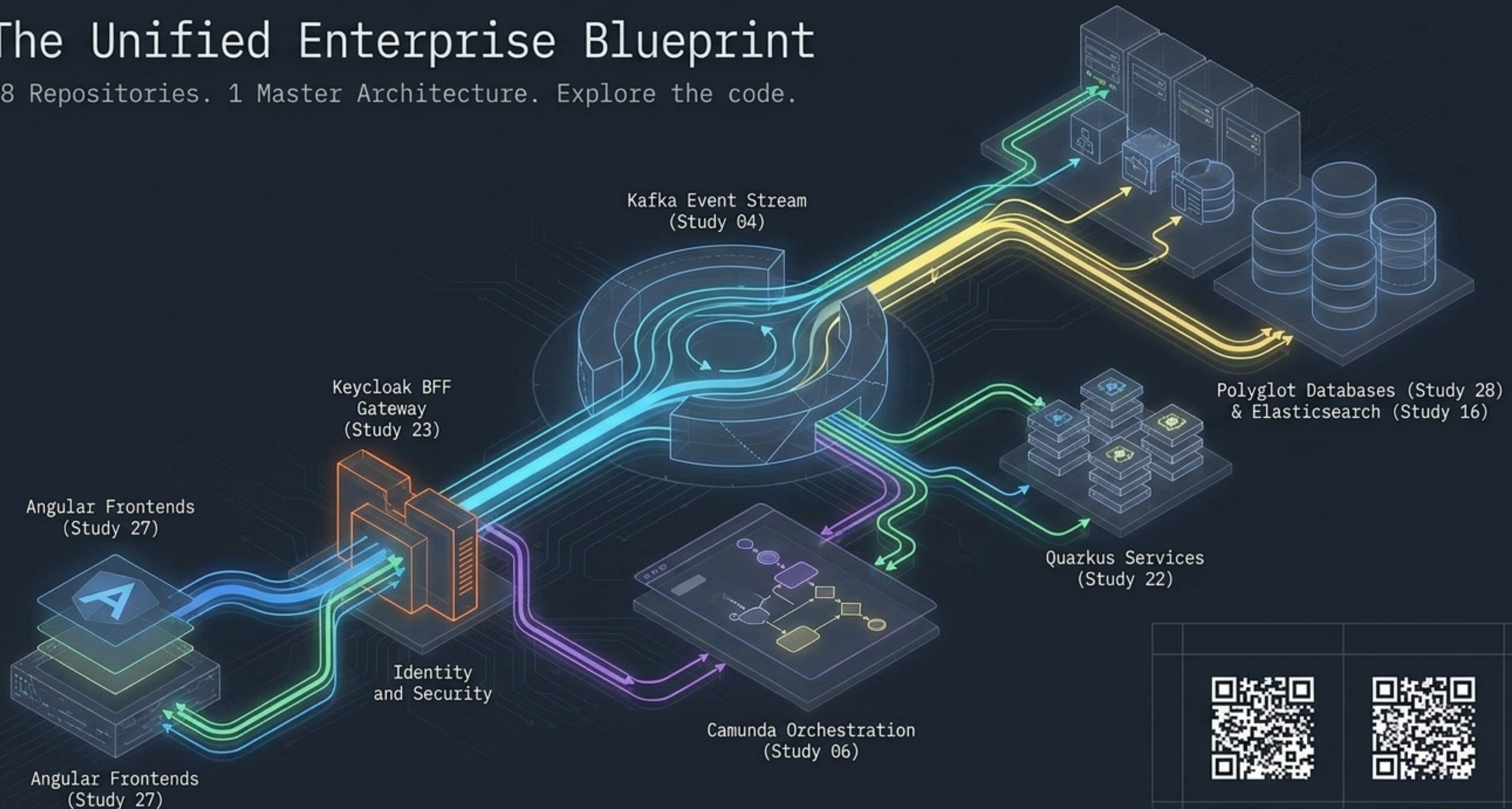
Modern Client Delivery

Decoupling frontend rendering from backend storage using Angular, Material Design, and Cypress E2E testing.



The Unified Enterprise Blueprint

28 Repositories. 1 Master Architecture. Explore the code.



k1729p/Miscellany



k1729p/Portfolio